

IDEAS-TIH SUMMER INTERNSHIP PROJECT REPORT

PROJECT TITLE:

Exploratory data analysis with Coffee sales data and MNIST data clustering.

Student Name:

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Course / Batch and Institute Name:

Bachelor in Computer Applications (BCA) at GITAM Deemed to be University

Period Of Internship

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Report Submitted To:

**IDEAS – Institute of Data Engineering, Analytics and Science Foundation,
ISI Kolkata**

1.Abstract:

The aim of this project was to explore and analyze a coffee sales dataset and apply clustering techniques to the MNIST handwritten digits dataset. The coffee sales data was examined to understand sales trends, customer preferences, and variations in revenue across different time periods. Various data preprocessing and analysis techniques were used to prepare and study the dataset. In addition, a small synthetic dataset was generated to understand the process of creating sample data for analysis. For the machine learning part of the project, the MNIST dataset was clustered using the K-Means algorithm. Since clustering does not directly produce actual class labels, a mapping technique was used to compare cluster assignments with the true digit labels. The performance of the clustering model was evaluated using the macro F1 score and a confusion matrix. The project provided practical exposure to data analysis, preprocessing, and unsupervised machine learning using Python and related libraries.

2.Introduction:

In today's data-driven world, organizations rely heavily on data analysis to understand patterns, improve decision-making, and gain valuable insights. Before applying advanced machine learning techniques, it is important to understand the structure and characteristics of the data through Exploratory Data Analysis (EDA). EDA helps identify trends, detect inconsistencies, and uncover useful information that may not be immediately visible.

This project consisted of two major tasks. The first task focused on analyzing a coffee sales dataset. The objective was to study sales behavior, identify popular coffee products, and understand how sales varied across different months, years, and times of the day. Through this analysis, meaningful business insights could be extracted from the data.

The second task involved working with the MNIST handwritten digits dataset, a widely used dataset in machine learning. K-Means clustering was applied to group similar digit images without using their actual labels during training. The

clustering results were then evaluated to understand how effectively the algorithm grouped the handwritten digits.

The entire project was carried out using Python in Google Colab with the help of libraries such as Pandas, NumPy, SciPy, and Scikit-Learn. Apart from learning technical concepts, the project also provided hands-on experience in organizing code, interpreting results, and documenting findings.

3.Project Objective:

The main objectives of this project were:

1. To perform exploratory data analysis on a real-world coffee sales dataset.
2. To identify sales patterns based on year, month, coffee type, and time of day.
3. To apply data preprocessing techniques and create useful features from existing data.
4. To generate a synthetic dataset for understanding data simulation concepts.
5. To implement and evaluate K-Means clustering on the MNIST handwritten digits dataset.
6. To assess clustering performance using F1 score and confusion matrix analysis.
7. To strengthen practical skills in data analysis and machine learning using Python.

4.Methodology:

Part A: Coffee Sales Data Analysis

Data Collection:

The coffee sales dataset was provided as part of the internship assignment. The dataset contained transaction-level information related to coffee sales.

Data Loading:

The dataset was imported into a Pandas DataFrame using the `read_csv()` function. This allowed efficient handling and analysis of the data.

Initial Data Inspection:

Before performing any analysis, the dataset was inspected to understand its structure and quality. The following checks were carried out:

- Number of rows and columns
- Duplicate records
- Missing values
- Overall dataset shape

Results obtained:

- Total Rows: 3547
- Total Columns: 11
- Duplicate Rows: 0
- Missing Values: 0

The dataset was found to be clean and did not require extensive cleaning.

Descriptive Statistics:

Basic statistical measures were generated using the `describe()` function. These statistics helped in understanding the distribution and range of numerical variables present in the dataset.

Data Preprocessing:

To make the dataset suitable for analysis, the following preprocessing steps were performed:

- Converted the Date column into datetime format.
- Extracted Month and Year information from the Date column.
- Verified the data types of all columns.

These newly created features were later used for aggregation and trend analysis.

Data Aggregation and Analysis:

Several analyses were performed to gain insights from the dataset:

- Average sales per year
- Maximum sales recorded in each month
- Total sales generated by each coffee type
- Identification of unique coffee varieties
- Average sales during different times of the day

These analyses helped in understanding customer preferences and sales behavior.

Part B: Synthetic Data Generation:

As part of the project, a synthetic dataset was created to understand how artificial data can be generated for testing and experimentation purposes.

The dataset consisted of:

- 100 sequential dates starting from January 1, 2023
- 100 randomly generated numerical values

The dataset was created using Pandas and NumPy functions and served as a simple example of data simulation.

Part C: MNIST Clustering Analysis:

1.Data Set Loading:

The MNIST handwritten digits dataset was loaded using the Scikit-Learn library.

2.Dataset dimensions:

- Features (X): 1797×64
- Labels (y): 1797

Each observation represented an image of a handwritten digit converted into numerical pixel values.

3.Model Selection:

K-Means clustering was selected for this task because it is one of the most commonly used unsupervised learning algorithms. Since the dataset contains ten-digit classes (0–9), the number of clusters was set to ten.

4.Model Training:

The clustering model was trained using the following parameters:

- Number of Clusters = 10
- Random State = 42

After training, cluster labels were assigned to all observations.

5.Label Mapping:

Since cluster labels generated by K-Means do not directly correspond to actual digit labels, each cluster was mapped to the most frequently occurring true label within that cluster. This allowed meaningful comparison between predicted and actual classes.

6.Performance Evaluation:

The clustering performance was evaluated using:

- Macro-Averaged F1 Score
- Confusion Matrix

7.GitHub Repository:

All project files, including the notebook and report, were uploaded to GitHub for documentation and version control.

GitHub Repository Link:

<https://github.com/UtpalaAditya/IDEAS-TIH-Summer-Internship-2026>

5.Data Analysis and Results:

Coffee Sales Dataset Results

1.Dataset Summary:

Metric	Value
Rows	3547
Columns	11
Duplicate Rows	0
Missing Values	0

The dataset was complete and did not contain any missing or duplicate records, making it suitable for analysis without additional cleaning.

2.Unique Coffee Types:

A total of 8 unique coffee products were identified in the dataset:

- Latte
- Hot Chocolate
- Americano
- Americano with Milk
- Cocoa
- Cortado
- Espresso

- Cappuccino

3.Sales Analysis:

The analysis revealed noticeable differences in sales across coffee types and time periods. Certain coffee varieties contributed more to overall revenue, while sales patterns also varied depending on the month and time of day.

Screenshot: Average Sales per Year

... Average Sales per Year:

	Year	money
0	2024	31.737634
1	2025	31.390011

Screenshot: Maximum Sales By Month:

Maximum Sales by Month Name:

	Month_name	money
0	Apr	38.70
1	Aug	32.82
2	Dec	35.76
3	Feb	35.76
4	Jan	35.76
5	Jul	37.72
6	Jun	37.72
7	Mar	38.70
8	May	37.72
9	Nov	35.76
10	Oct	35.76
11	Sep	35.76

Screenshot: Total Sales by Coffee Type:

*** Total Sales by Coffee Name:

	coffee_name	money
0	Americano	14650.26
1	Americano with Milk	24751.12
2	Cappuccino	17439.14
3	Cocoa	8521.16
4	Cortado	7384.86
5	Espresso	2690.28
6	Hot Chocolate	9933.46
7	Latte	26875.30

Screenshot: Number Of Unique Coffee Names and Their Names:

Number of Unique Coffee Types:

8

Unique Coffee Names:

['Latte', 'Hot Chocolate', 'Americano', 'Americano with Milk', 'Cocoa', 'Cortado', 'Espresso', 'Cappuccino']

Screenshot: Average Sales by Time of Day:

Average Sales by Time of Day:

	Time_of_Day	money
0	Afternoon	31.643187
1	Morning	30.422693
2	Night	32.890904

4.MNIST Clustering Results:

Dataset Shape:

Dataset	Shape
Features (X)	(1797, 64)
Labels (y)	(1797,)

5.Clustering Performance:

Metric	Value
Number of Clusters	10
F1 Score	0.8628

The clustering model achieved a macro F1 score of approximately **0.86**, indicating that the clusters formed by the algorithm aligned well with the actual digit classes. Considering that K-Means is an unsupervised learning algorithm, this result demonstrates reasonably strong clustering performance.

6.Confusion Matrix

The confusion matrix was generated to compare the mapped cluster labels with the actual digit labels. It provided a detailed view of how accurately each digit class was grouped by the clustering algorithm.

Screenshot: Confusion Matrix Output

	0	1	2	3	4	5	6	7	8	9
0	176	0	0	0	2	0	0	0	0	0
1	0	130	23	1	0	1	2	0	3	22
2	1	6	146	12	0	0	0	2	9	1
3	0	0	1	159	0	2	0	7	6	8
4	0	13	0	0	159	0	0	5	4	0
5	0	0	0	0	2	137	2	0	0	41
6	1	3	0	0	0	0	176	0	1	0
7	0	0	0	0	0	0	0	169	10	0
8	0	14	3	0	0	2	1	2	136	16
9	0	1	0	1	3	4	0	8	3	160

6.Conclusion:

This project provided valuable practical experience in both exploratory data analysis and machine learning. Through the analysis of the coffee sales dataset, it was possible to understand sales trends, identify popular coffee products, and observe how sales varied across different periods. The dataset

was clean and well-structured, which allowed the analysis to be carried out efficiently.

The machine learning component introduced the application of K-Means clustering on the MNIST handwritten digits dataset. After mapping cluster labels to actual digit classes, the model achieved a macro F1 score of 0.8628, indicating that the clustering algorithm was able to group similar digit images effectively. The confusion matrix further supported these findings by showing a strong correspondence between clusters and actual labels.

Overall, this project strengthened my understanding of data preprocessing, exploratory analysis, feature engineering, clustering techniques, and model evaluation. It also provided hands-on experience with Python libraries and GitHub-based project management. In future work, additional clustering algorithms and visualization techniques could be explored to further improve performance and gain deeper insights.

7.APPENDICES:

Appendix A: References:

- 1.IDEAS TIH SUMMER INTERNSHIP LIVE CLASSES
- 2.RECORDED LECTURES PROVIDED
- 3.GOOGLE COLAB NOTES PROVIDED
- 4.GOOGLE CHROME FOR DOUBTS

Appendix B: GitHub Repository:

GitHub Link:

<https://github.com/UtpalaAditya/IDEAS-TIH-Summer-Internship-2026>

Appendix C: Project Files:

- Project Notebook (.ipynb)
- Project Report (PDF)

Appendix D: Additional Documents:

A Recorded Video of my PROJECT explanation will be submitted in the google form link provided by the IDEAS-TIH 4_WEEK SUMMER INTERNSHIP TEAM.

**Thank you for offering me the internship opportunity at
IDEAS-TIH, Kolkata**